

MIT Open Learning Library — Standard Course Extraction

MITx 8.370.1x — SU2: Quantum mechanics: qubits and tensor products

Populated module/evidence-package record for QAI Enterprise Library extraction.

Evidence boundary: MIT course evidence supplied through the course structure, screenshots, solved questions and discussion is separated from QAI interpretation. Primitive candidates are not production claims.

1. Course Identity

Field	Captured information
Course title	Quantum mechanics: qubits and tensor products
MITx / course ID	MITx 8.370.1x
Term / course version	1T2018
MIT Open Learning Library URL	https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/
Primary domain	Quantum mechanics for computation; qubits; quantum states; unitary transformations; measurement; tensor products.
Course purpose	SU2 develops the mathematical and operational foundations for single-qubit quantum mechanics, measurement, unitary operators, basis changes, rotations and tensor products of states/operators.
Source material available	SU2 hierarchy supplied by the user; previously solved concept questions; SU2 PS#2 screenshots and accepted solutions; mathematical expressions/hints shown in screenshots; reasoning captured in this session.
Extraction status	Complete for the supplied SU2 evidence package. A complete independent page-by-page crawl, exact total question count and notebooks were not supplied.

2. Course Fidelity and Source Structure

- Subunit: SU2: Quantum mechanics: qubits and tensor products.
- Lecture U1.3: Quantum Mechanics I – Qubits.
- Lecture U1.4: Quantum Mechanics II – measurement and tensor products.
- Previously captured concept questions retained as evidence: Equivalent quantum states; Bloch sphere states; Valid unitary matrices; Rotation axes of a Hadamard gate; Measurement outcome probabilities; Tensor product of X and Y; Matrix description of quantum circuit.
- Problem set: PS#2 - Quantum Mechanics.
- Supplied PS#2 evidence includes Pauli matrices/eigenvalues and eigenstates; basis changes; quantum gates; single-qubit rotation/Bloch sphere; tensor products of states; tensor products of matrices/CNOT and conjugation.
- No executable notebook/IPYNB was supplied for SU2.

3. Course Map

Subunit	Item type	Questions / evidence	Source available	Status
SU2	Lecture U1.3	Previously captured CQs on equivalent states, Bloch sphere, unitary matrices, Hadamard rotation	Yes	Captured

		axis		
SU2	Lecture U1.4	Previously captured CQs on measurement probabilities, tensor product, matrix circuit description	Yes	Captured
SU2	PS#2 - Quantum Mechanics	Supplied solved PS screenshots covering Pauli matrices, basis changes, quantum gates, rotations, tensor products and CNOT-related transformations	Yes	Complete for supplied evidence
SU2	Concept questions	7 previously analyzed CQs explicitly retained	Yes	Captured
SU2	Notebook / executable assets	None supplied	No	Pending

4. Module / Subunit Record

Field	SU2 record
Subunit ID / name	SU2: Quantum mechanics: qubits and tensor products
MIT source URL	https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/#block-v1:MITx+8.370.1x+1T2018+type@chapter+block@Week2
Learning objectives	Develop understanding of qubit states, equivalent state representations, Bloch-sphere geometry, unitary matrices, quantum measurement, Pauli operators, basis changes, quantum gates and tensor products of states and matrices.
Lectures / topics	U1.3: Quantum Mechanics I – Qubits; U1.4: Quantum Mechanics II – measurement and tensor products.
Problem sets	PS#2 - Quantum Mechanics.
Questions	Previously captured CQs plus the supplied PS#2 solved questions. Exact complete MIT question count is not independently asserted.
Important concepts	Global phase; Bloch sphere; unitary condition; eigenvalues/eigenstates; Pauli matrices; basis transformation; phase gate; gate composition; axis-angle rotations; tensor products; CNOT; controlled operators; operator conjugation.
References	No additional external reference was newly supplied for SU2.

5. Lecture / Content Records

U1.3 — Quantum Mechanics I – Qubits

Field	Captured record
Type	Lecture / concept content
MIT source URL	https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/#block-v1:MITx+8.370.1x+1T2018+type@sequential+block@lectures_u1_3
Question count	Seven previously captured concept questions are retained; complete lecture count not independently supplied.
Concepts covered	Equivalent quantum states/global phase; Bloch sphere; valid

	unitaries; Hadamard rotation axis; Pauli/eigenstate geometry; single-qubit state parameterization.
Key mathematical ideas	$ \psi\rangle = \cos(\theta/2) 0\rangle + e^{i\phi}\sin(\theta/2) 1\rangle$; $U^\dagger U = I$; eigenvalue/eigenstate relation; Bloch coordinates; axis-angle interpretation of single-qubit unitaries.
Implementation relevance	State representation, unitary validation, eigenstructure analysis, basis conversion, Bloch visualization and operator representation.

U1.4 — Quantum Mechanics II – measurement and tensor products

Field	Captured record
Type	Lecture / concept content
MIT source URL	https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/#block-v1:MITx+8.370.1x+1T2018+type@sequential+block@lectures_u1_4
Question count	Three previously captured concept questions are retained; complete lecture count not independently supplied.
Concepts covered	Measurement outcome probabilities; tensor products of operators; matrix description of quantum circuits; composite operator representation.
Key mathematical ideas	Born-rule probability $P(i) = \alpha_i ^2$; tensor/Kronecker product; block-matrix structure; circuit unitary composition.
Implementation relevance	Composite state/operator construction, controlled operations, structured operator representation, circuit-to-matrix translation.

6. Problem Set Record

Field	PS#2 record
Problem set ID / title	PS#2 - Quantum Mechanics
MIT source URL	https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/#block-v1:MITx+8.370.1x+1T2018+type@sequential+block@ps2
Question count	Exact complete PS count not independently enumerated; supplied screenshots cover the solved PS items listed below.
Original problem available	Yes — problem statements and mathematical instructions visible in supplied screenshots.
Solved questions available	Yes — supplied screenshots show accepted/correct solutions.
Notebook / code available	No notebook supplied; answers were entered into the MIT OLL problem interface.
Related lectures	U1.3 and U1.4.

7. Solved Question Extraction

PS2-Q1 — Pauli matrices — Eigenvalues, eigenstates

Field	Record
Problem statement / required result	Compute positive/negative eigenvalues and eigenstates for X, Y, Z .
Solution / reasoning	Solve $A \psi\rangle = \lambda \psi\rangle$; identify normalized eigenvectors in the $\{ 0\rangle, 1\rangle\}$ basis.
Mathematical derivation / result	$X: +1(0\rangle + 1\rangle)/\sqrt{2}, -1(0\rangle - 1\rangle)/\sqrt{2}$. $Y: +1(0\rangle + i 1\rangle)/\sqrt{2}, -1(0\rangle - i 1\rangle)/\sqrt{2}$. $Z: +1 0\rangle, -1 1\rangle$.
Key insight	Eigenvectors define natural measurement/representation bases of operators.
Validation / result	Accepted/correct in supplied screenshot.

PS2-Q2 — Basis changes

Field	Record
Problem statement / required result	Express Pauli matrices in the $\{ +\rangle, -\rangle\}$ basis defined by H.
Solution / reasoning	Use $A_{\text{new}} = S^\dagger A_{\text{old}} S$ with $S=H$ and $H^\dagger=H$.
Mathematical derivation / result	$X \rightarrow Z, Y \rightarrow -Y, Z \rightarrow X$ in the supplied basis representation.
Key insight	Operator representation depends on basis; the underlying operation is not thereby changed.
Validation / result	Accepted/correct in supplied screenshot.

PS2-Q3 — Quantum gates

Field	Record
Problem statement / required result	Introduce phase gate S and compute HX, SH and $SXS^\dagger$.
Solution / reasoning	Sequential gates compose by matrix multiplication; rightmost operation acts first; conjugation transforms operators.
Mathematical derivation / result	$HX = (1/\sqrt{2})[[1,1],[-1,1]]$; $SH = (1/\sqrt{2})[[1,1],[i,-i]]$; $SXS^\dagger = Y$.
Key insight	Gate sequence becomes a composite unitary; conjugation provides an operator-transformation mechanism.
Validation / result	3/3 points shown.

PS2-Q4 — Single qubit rotation

Field	Record
Problem statement / required result	Apply S, SH and SHSH to the state at $(\pi/2, 0)$ and express results on Bloch sphere.
Solution / reasoning	Use the state parameterization and gate rotation interpretation.
Mathematical derivation / result	$S \rightarrow (\pi/2, \pi/2)$; $SH \rightarrow (0, 0)$; $SHSH \rightarrow (\pi/2, \pi/2)$.
Key insight	Same transformation can be interpreted algebraically as matrices or geometrically as Bloch-sphere rotations.
Validation / result	3/3 points shown.

PS2-Q5 — Single qubit rotation — axis/angle

Field	Record
Problem statement / required result	Identify rotation axis and angle for X, H and S.
Solution / reasoning	Use $U = e^{-i\theta \mathbf{n} \cdot \boldsymbol{\sigma}/2} = \cos(\theta/2)I - i \sin(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}$ up to global phase.
Mathematical derivation / result	X: $\mathbf{n}=(1,0,0)$, $\theta=\pi$. H: $\mathbf{n}=(1/\sqrt{2}, 0, 1/\sqrt{2})$, $\theta=\pi$. S: $\mathbf{n}=(0,0,1)$, $\theta=\pi/2$.
Key insight	Unitary matrices admit geometric axis-angle representation for single qubits.
Validation / result	All displayed entries marked correct.

PS2-Q6 — Tensor products of states

Field	Record
Problem statement / required result	Construct composite state vectors from tensor products.
Solution / reasoning	Multiply amplitudes pairwise and preserve tensor ordering.
Mathematical derivation / result	$ 00\rangle = [1, 0, 0, 0]^T$; $ ++\rangle = [1/2, 1/2, 1/2, 1/2]^T$.
Key insight	Tensoring expands subsystem state spaces from dimensions n, m to nm .
Validation / result	2/2 points shown.

PS2-Q7 — Tensor products of matrices / CNOT

Field	Record
Problem statement / required result	Compute $I \otimes X$ and $(I \otimes H) U_{\text{CNOT}} (I \otimes H)$, using block tensor construction.

Solution / reasoning	Construct $A \otimes B$ as blocks $A_{ij} B$; represent CNOT as $ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes X$; apply conjugation.
Mathematical derivation / result	$I \otimes X$ is a 4×4 block-structured operator; the conjugated CNOT result shown is $\text{diag}(1, 1, 1, -1)$, i.e. the controlled-Z matrix.
Key insight	Tensor products create structured composite operators; controlled operations can be represented using projector-selected operators; conjugation can transform gate families.
Validation / result	2/2 points shown.

Previously captured concept questions

Question	Evidence retained	QAI-relevant interpretation
Equivalent quantum states	All displayed states equivalent up to global phase.	State equivalence / canonical representation.
Bloch sphere states	$ 0\rangle$ at $+z$; $(0\rangle + 1\rangle)/\sqrt{2}$ at $+x$; $(0\rangle + i 1\rangle)/\sqrt{2}$ at $+y$; opposite-axis states supplied.	Geometric state representation.
Valid unitary matrices	Valid matrices satisfy $U^\dagger U = I$; screenshots showed accepted selections.	Unitary validation.
Rotation axes of a Hadamard gate	Axis point $(1/\sqrt{2}, 0, 1/\sqrt{2})$.	Eigenstate geometry / axis-angle representation.
Measurement outcome probabilities	Equal probabilities arise from equal amplitude magnitudes for selected outcomes.	Born-rule measurement interface.
Tensor product of X and Y	$X \otimes Y$ forms a 4×4 operator with block structure.	Composite operator construction.
Matrix description of quantum circuit	Circuit represented by product of gate matrices in execution order.	Circuit $\leftrightarrow$ operator $\leftrightarrow$ matrix translation.

8. Notebook / Computational Asset Record

Field	Captured record
Notebook name	Not supplied.
Course / problem-set relationship	PS#2 answers entered directly into the MIT OLL problem interface.
Original source	MIT OLL SU2 lectures and PS#2 screenshots supplied in this conversation.
Language / framework / SDK	MIT course answer-entry syntax; no general-purpose SDK supplied.
Dependencies	Not supplied.
Input	Matrices, state vectors, basis definitions, Bloch coordinates and gate sequences specified by the PS.
Output	Eigenstructures, transformed matrices, Bloch coordinates and tensor-product matrices.
Algorithm / model	Linear algebra, eigenvalue/eigenvector analysis, basis transformation, unitary composition, axis-angle representation and tensor products.
Execution environment	MIT OLL course interface as evidenced by supplied screenshots.
Result	Displayed PS answers were accepted/correct.
Validation	Course grading/correctness indicators and rendered mathematical results.

9. Algorithms and Mathematical Models

Model / algorithm	Mathematical form	Inputs	Outputs	Evidence
Pure single-qubit state	$ \psi\rangle = \cos(\theta/2) 0\rangle + e^{i\phi}\sin(\theta/2) 1\rangle$	θ, ϕ	State vector / Bloch position	Lecture hint / CQ
Unitary condition	$U^\dagger U = I$	U	Validity / inverse $U^\dagger$	CQ
Eigenstructure	$A \psi\rangle = \lambda \psi\rangle$	Operator A	Eigenvalues/eigenvectors	PS2-Q1
Basis transformation	$A_{\text{new}} = S^\dagger A_{\text{old}} S$	A, S	Operator in new basis	PS2-Q2
Gate composition	$U_{\text{total}} = U_m \dots U_2 U_1$	Gate matrices	Composite unitary	PS2-Q3
Operator conjugation	$A' = U A U^\dagger$	Operator A, unitary U	Transformed operator	PS2-Q3/Q7
Axis-angle unitary	$U = e^{-i\theta \mathbf{n} \cdot \boldsymbol{\sigma}/2} = \cos(\theta/2)I - i \sin(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}$	$\mathbf{n}, \theta$	Single-qubit unitary	PS2-Q5
Born-rule measurement	$P(i) = \alpha_i ^2$	State amplitudes	Outcome probabilities	Previously captured CQ
Tensor product of states	$ a\rangle \otimes b\rangle$	Subsystem state vectors	Composite vector	PS2-Q6
Tensor product of matrices	$A \otimes B = [A_{ij} B] \text{ block matrix}$	A, B	Composite operator	PS2-Q7
Controlled operator	$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes X$	Control projectors, target operators	CNOT operator	PS2-Q7
Circuit matrix representation	$U_C = U_m \dots U_1$	Gate sequence	Complete unitary	Previously captured CQ / PS2

10. Extracted Patterns

Pattern	Source evidence	Problem-solving structure	Inputs → outputs	Control / recurrence	Evidence level	Potential reuse
Eigenstructure → Natural Representation	PS2-Q1/Q2	Extract eigenvalues/eigenvectors and use the eigenbasis as a natural representation.	Operator → eigenvalues/eigenvectors/basis	Basis selection; repeated within PS	Moderate / direct PS evidence	Operator-analysis and representation service.
Basis / Representation Transformation	Previously captured CQs + PS2-Q2	Transform operator or state representations while preserving mathematical semantics.	State/operator + basis → equivalent representation	Basis selection; repeated	Strong within SU2	QAI intermediate-representation and conversion layer.
Gate → Composite Unitary	PS2-Q3 + prior CQ	Compose sequential gate operators into a single unitary.	Gate sequence → composite unitary	Execution order; repeated	Strong	Circuit compiler/runtime representation.
Unitary ↔ Axis-Angle / Bloch Geometry	PS2-Q4/Q5 + prior CQ	Represent a single-qubit unitary geometrically as an axis-angle rotation and state evolution on the Bloch sphere.	Single-qubit U/state → geometric representation	Rotation axis/angle	Strong SU2 evidence	Visualization, control, optimization and operator-analysis.
Tensor State / Operator Composition	PS2-Q6/Q7 + prior CQ	Construct composite states/operators from subsystem factors using tensor products.	Subsystem states/operators → composite state/operator	Tensor/block structure; repeated	Strong	Tensor-backed state/operator layer.
Projector-	PS2-Q7	Build controlled	Projectors + target	Conditiona	Direct	Controlled-

Controlled Operator Construction		operations from projectors and subsystem operators.	operators → controlled operator	l execution	PS evidence	operation library/runtime .
Structured Block / Tensor Representation	PS2-Q7 + QAI-derived analysis	Preserve block structure rather than treating expanded matrices as undifferentiated data.	Composite operator → structured operator	Block/factor metadata	Course evidence + QAI inference	Future structured operator analyzer / scoped computation.

11. Primitive Candidates

Eigenstructure Analyzer

Field	Candidate record
Source evidence	PS2-Q1/Q2
Pattern	Extract eigenvalues/eigenvectors and expose a natural operator basis.
Proposed inputs	Operator
Proposed outputs	Eigenvalues/eigenvectors + basis metadata
API concept	<code>eigen_decompose(A)</code>
Runtime requirements	Linear-algebra runtime; operator model
Validation required	Eigenpair residuals and reconstruction $A \approx \text{SDS}^\dagger$
Maturity	Candidate / future
Promotion recommendation	Do not promote from one course; require independent and cross-course validation.

Basis Transform

Field	Candidate record
Source evidence	PS2-Q2
Pattern	Transform state/operator representations between bases.
Proposed inputs	Operator/state + basis transform
Proposed outputs	Equivalent representation
API concept	<code>basis_transform(A,S) / state_transform(ψ>,S)</code>
Runtime requirements	State/operator model
Validation required	Round-trip transformation and norm/unitarity checks
Maturity	Candidate / future
Promotion recommendation	Validate representation equivalence and numerical stability.

Unitary Composition

Field	Candidate record
Source evidence	PS2-Q3 + prior CQ
Pattern	Compose sequential quantum gates/operators.
Proposed inputs	Ordered unitary list
Proposed outputs	Composite unitary
API concept	<code>compose_unitaries(U1,...,Um)</code>
Runtime requirements	Operator model / circuit IR
Validation required	Compare matrix product with circuit execution
Maturity	Candidate / future
Promotion recommendation	Strong repeated evidence, but no production promotion yet.

Operator Conjugation

Field	Candidate record
Source evidence	PS2-Q3/Q7

Pattern	Apply UAU [†] to transform an operator.
Proposed inputs	Operator A, unitary U
Proposed outputs	Transformed operator
API concept	conjugate_operator(U,A)
Runtime requirements	Operator model
Validation required	Algebraic equivalence and unitary checks
Maturity	Candidate / future
Promotion recommendation	Validate against Pauli/CNOT transformations.

Bloch / Axis-Angle Conversion

Field	Candidate record
Source evidence	PS2-Q4/Q5 + prior CQ
Pattern	Convert between single-qubit matrix, axis-angle and Bloch representations.
Proposed inputs	Single-qubit U/state
Proposed outputs	Axis-angle / Bloch representation
API concept	unitary_to_axis_angle(), state_to_bloch()
Runtime requirements	Single-qubit state/operator model
Validation required	Round-trip equivalence and Bloch rotation tests
Maturity	Candidate / future
Promotion recommendation	Validate numerical edge cases and global-phase handling.

Tensor Composition

Field	Candidate record
Source evidence	PS2-Q6/Q7 + prior CQ
Pattern	Construct composite state/operator using tensor products.
Proposed inputs	Subsystem states/operators
Proposed outputs	Composite vector/matrix
API concept	tensor_state(), tensor_operator()
Runtime requirements	Tensor backend / state model
Validation required	Dimension/order checks and algebraic equivalence
Maturity	Candidate / future
Promotion recommendation	Cross-course tensor validation required.

Controlled Operator Builder

Field	Candidate record
Source evidence	PS2-Q7
Pattern	Construct controlled operations from projectors and target operators.
Proposed inputs	Control projectors + target operators
Proposed outputs	Controlled unitary
API concept	controlled_operator(P_i,U_i)
Runtime requirements	Operator/circuit runtime
Validation required	Truth tables and full-matrix equivalence
Maturity	Candidate / future
Promotion recommendation	Validate multi-control/generalized forms.

Structured Tensor / Block Analyzer

Field	Candidate record
Source evidence	PS2-Q7 + QAI-derived analysis
Pattern	Inspect tensor/block structure and optionally scope computation to factors/blocks.
Proposed inputs	Structured composite operator
Proposed outputs	Factors, blocks, scoped representation
API concept	analyze_structure(), scope_subsystem()
Runtime requirements	Structured tensor/operator runtime

Validation required	Benchmark against full-matrix execution and verify exact equivalence
Maturity	Future capability
Promotion recommendation	QAI inference; requires independent structured-operator/tensor-network validation.

12. QAI Architecture Mapping

QAI layer	SU2 mapping
QAI Language	Represent qubit states, Pauli operators, gates, unitary operators, basis transforms, tensor products and Bloch/axis-angle forms as explicit semantic constructs.
QAI Primitives	Candidate abstractions: eigenstructure analysis, basis transform, unitary composition, conjugation, Bloch/axis-angle conversion, tensor composition and controlled-operator construction.
QAI OS	Provide logical state/operator abstractions and representation metadata; SU2 is foundational evidence, not a complete OS design.
QAI Runtime	Execute/combine single- and multi-qubit operators, state transformations and tensor compositions.
Hybrid Runtime	Potential classical coordination for representation analysis, validation, visualization and execution-representation selection; not directly implemented by SU2.
QAI Control Plane	Potential management of representation selection, circuit/operator transformation policies and resource-aware decomposition; future mapping.
Quantum Control Plane	Potential management of single-qubit rotations, basis selection, state preparation and measurement; not directly implemented.
Adaptive Network Fabric	Not directly evidenced by SU2; future mapping only.
Resource Registry	Potential registry of qubit dimensions, operator sizes, tensor factors and execution-resource requirements.
Capability Registry	Candidate capabilities: eigenstructure analysis, unitary composition, basis transformation, tensor composition, controlled operators and Bloch/axis-angle conversion.
QAI Labs	Natural validation location for single-qubit state/operator models, tensor composition, circuit equivalence and representation round trips.
QAI Product Foundry	Potential source for reusable validated quantum/operator components after independent validation; not a direct SU2 deliverable.

13. Classification

- Educational knowledge — qubit representations, Pauli matrices, quantum gates, Bloch sphere, measurement and tensor products.
- Mathematical model — eigenstructure, unitary matrices, basis transformation, axis-angle representation, Born rule and tensor products.
- Algorithm / procedure — matrix composition, basis transformation, eigenvector extraction and tensor-product construction.
- Pattern — operator representation transformation; gate composition; tensor/block composition; projector-controlled operations.
- Implementation evidence — supplied PS#2 answers were accepted/correct in the MIT OLL interface.
- Reusable capability — candidate only; independent implementation and validation are still required.
- Future capability — structured tensor/block analysis and scoped computation remain QAI-derived hypotheses.

- Research — not claimed from SU2 alone.

14. Evidence and Provenance

Field	Evidence
MIT OLL source URL(s)	Course: https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/ ; SU2 Week 2: https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/#block-v1:MITx+8.370.1x+1T2018+type@chapter+block@Week2 ; U1.3/U1.4 and PS#2 URLs supplied in the conversation.
Original document / problem-set reference	MITx 8.370.1x, SU2, U1.3, U1.4, PS#2 - Quantum Mechanics.
Notebook reference	Not supplied.
Solution reference	Screenshots supplied in this conversation and associated reasoning/solution discussion.
External reference	None newly supplied for SU2.
Evidence limitations	Based on supplied course structure, screenshots, hints and accepted answers; not a complete independent crawl of every MIT OLL page. Exact complete question counts and notebook inventory are not asserted.

15. Cross-Course Convergence

Course / topic	Common pattern	Primitive candidate	Evidence status
SU1: Quantum and classical computing fundamentals	State/operator representation; tensor composition; circuit $\leftrightarrow$ matrix	State representation; tensor composition; circuit/operator/matrix translation	Internal recurrence strengthened by SU2.
SU1: reversible circuits	Composition/decomposition and operator/circuit representations	Composition; structured representation	Strong related evidence; SU2 adds quantum operator/eigenbasis geometry.
MIT 8.371.3x — prior extracted HSP/Shor material	Quantum operator, controlled operation, measurement and estimation	AQSE and operator/control abstractions	Related concepts; cross-course validation not yet completed.
Future MIT courses	Eigenstructure, tensor methods, operator transformations, quantum algorithms	To be determined	Pending future extraction.

Cross-course convergence is not treated as fully validated until the other course evidence packages are extracted and compared.

16. Reuse and Implementation Assessment

- Direct reuse: MIT examples can serve as reference test vectors and educational examples; not production code.
- Pattern reuse: strong opportunity for operator composition, basis transformation, eigenstructure, tensor composition and controlled-operator construction.
- Design reference: strong for a representation layer connecting state, operator, matrix, circuit, basis and geometric forms.
- Research input: structured tensor/block analysis and scoped computation are promising QAI directions but remain hypotheses.
- Architecture input: high — SU2 reinforces a layered representation and operator-algebra architecture.
- Prototype candidate: single-qubit operator/state library, tensor operator builder and circuit/operator round-trip validator.

- New implementation required: QAI-native APIs, validation harness, runtime integration, structured representation analysis and telemetry.
- Historical / educational reference only: course-history/context material unless reused for Academy/knowledge assets.

17. Validation Plan

Validation area	Plan
Reference implementation needed	Yes — implement Pauli matrices, eigenstructure, basis transforms, phase/H/X/Z gates, axis-angle conversion, tensor products and CNOT/CZ equivalence in the QAI lab.
Simulator validation	Validate normalization, $U^\dagger U = I$, eigenvector equations, basis round trips, Bloch rotations, tensor dimensions, CNOT/CZ equivalence and measurement predictions.
Hardware / QPU validation	Optional later stage for physical single-/multi-qubit gate behavior; not required to establish the mathematical primitives.
Independent problem validation	Add random/unit-test matrices, alternative bases, additional controlled gates, multi-qubit tensor cases and independent circuit-equivalence tests.
Cross-course validation	Compare with later MIT courses covering algorithms, estimation, tensor networks, QEC and quantum control.
Benchmark	Compare dense-matrix and structured/tensor implementations for correctness, memory, execution time and scaling.
Acceptance criteria	Exact/within-tolerance equivalence; eigenpair residuals below threshold; unitary preservation; norm/probability preservation; tensor dimensions/order correct; circuit/operator round trips consistent; controlled-gate truth tables correct.
Promotion gate	No production primitive promotion from SU2 alone. Follow Candidate → Specification → Reference Implementation → Validation → Reusable Primitive → Enterprise Library.

18. Final Course Summary

- Major concepts: qubit state representation; global phase; Bloch sphere; unitary matrices; Pauli matrices; eigenvalues/eigenstates; basis changes; phase and Hadamard gates; single-qubit rotations; measurement; tensor products; controlled operators; CNOT/CZ relationship.
- Important algorithms/procedures: eigenstructure calculation; basis transformation $A' = S^\dagger A S$; sequential gate multiplication; operator conjugation $U A U^\dagger$; axis-angle extraction; tensor-product construction; projector-controlled operator construction.
- Important problem-solving patterns: choose a natural basis; transform representations without changing semantics; compose operations into a unitary; interpret unitary action geometrically; construct composite systems/operators from tensor factors; express conditional operations using projectors.
- Important solved questions: seven previously captured SU2 concept questions plus supplied PS#2 items on Pauli matrices, basis changes, quantum gates, single-qubit rotations, tensor states and tensor operators/CNOT.
- Important notebooks: None supplied.
- Primitive candidates: Eigenstructure Analyzer; Basis Transform; Unitary Composition; Operator Conjugation; Bloch/Axis-Angle Conversion; Tensor Composition; Controlled Operator Builder; Structured Tensor/Block Analyzer.
- Enterprise Library contributions: candidate abstractions for quantum state/operator representation, eigenstructure, basis transformation, geometric single-qubit representation, tensor composition and controlled operators.
- QAI architecture contributions: reinforces a representation-transform layer and operator-algebra layer between QAI language semantics and runtime execution; provides a foundation for tensor-backed and geometry-aware execution.

- Implementation opportunities: QAI single-qubit state/operator library; representation IR; tensor operator builder; controlled-gate library; Bloch/axis-angle visualization and conversion service; QAI Lab validation suite.
- Evidence gaps: full independent course crawl, exact complete question counts, lecture transcripts, notebooks and independent implementation/hardware evidence are not supplied.

19. Extraction Status

Area	Status	Notes
Course structure	Complete for supplied evidence	SU2 hierarchy and U1.3/U1.4/PS#2 captured.
Modules / subunits	Complete	SU2 captured.
Lectures	Captured	U1.3 and U1.4 identified; full lecture-page inventory pending.
Problem sets	Complete for supplied evidence	PS#2 solved items captured from screenshots.
Solved questions	Complete for supplied evidence	Previously captured CQs plus PS#2 items summarized.
Notebooks	Pending	None supplied.
Algorithms	Captured	Eigenstructure, basis transforms, unitary composition, axis-angle, tensor construction and controlled operators.
Mathematical models	Captured	Qubit parameterization, Bloch sphere, unitarity, eigenstructure, tensor products, Born rule and operator conjugation.
Patterns	Captured	Recurring representation/composition patterns identified; QAI-derived structured scoping kept separate.
Primitive candidates	Candidate	No production promotion.
QAI mapping	Captured	Architecture relationships recorded.
Cross-course mapping	Pending	SU1 recurrence identified; broader convergence awaits later course extraction.
Validation	Planned	Reference implementation, simulator, independent and later hardware validation planned.
Final review	Complete for supplied evidence	SU2 module/evidence package consolidated.

20. Extraction Rules

- Preserve the actual MIT OLL hierarchy and terminology.
- Keep original problem statements and solved questions associated.
- Treat solved questions as first-class evidence.
- Treat notebooks as implementation evidence when supplied.
- Do not infer implementation maturity from educational material.
- Separate MIT source evidence from QAI interpretation.
- Promote patterns to primitive candidates only when supported by evidence.
- Promote primitives toward the Enterprise Library only after appropriate validation.
- Use cross-course recurrence to strengthen, not replace, validation.
- Master working rule: PRESERVE → EXTRACT → ANALYZE → GENERALIZE → VALIDATE → MAP → PROMOTE.

QAI Retention Snapshot

SU2 should be retained as a foundational evidence package for the QAI stack. Its strongest implementation value is the relationship between quantum state representation, operator/eigenstructure, basis transformation, geometric single-qubit representation, gate composition, tensor composition and controlled operators. Together with SU1, it strengthens the case for a layered QAI representation and operator-algebra model. Structured tensor/block analysis and scoped computation remain QAI-derived future capabilities until independently validated.